

Trigonometry Identities Cheat Sheet

Complete trigonometry identities cheat sheet with Pythagorean, reciprocal, quotient, double angle, half angle, and sum/difference formulas. Free PDF download.

Pythagorean Identities

NAME	FORMULA	NOTES
Basic	$\sin^2(\theta) + \cos^2(\theta) = 1$	The fundamental identity
Tangent Form	$1 + \tan^2(\theta) = \sec^2(\theta)$	Divide by $\cos^2$
Cotangent Form	$1 + \cot^2(\theta) = \csc^2(\theta)$	Divide by $\sin^2$

Reciprocal Identities

NAME	FORMULA	NOTES
Cosecant	$\csc(\theta) = \frac{1}{\sin(\theta)}$	
Secant	$\sec(\theta) = \frac{1}{\cos(\theta)}$	
Cotangent	$\cot(\theta) = \frac{1}{\tan(\theta)}$	

Quotient Identities

NAME	FORMULA	NOTES
Tangent	$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$	
Cotangent	$\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$	

Double Angle Formulas

NAME	FORMULA	NOTES
Sine	$\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$	
Cosine (form 1)	$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$	
Cosine (form 2)	$\cos(2\theta) = 2 \cos^2(\theta) - 1$	
Cosine (form 3)	$\cos(2\theta) = 1 - 2 \sin^2(\theta)$	
Tangent	$\tan(2\theta) = \frac{2 \tan(\theta)}{1 - \tan^2(\theta)}$	

Half Angle Formulas

NAME	FORMULA	NOTES
Sine	$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos(\theta)}{2}}$	
Cosine	$\cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos(\theta)}{2}}$	
Tangent	$\tan\left(\frac{\theta}{2}\right) = \frac{1 - \cos(\theta)}{\sin(\theta)}$	

Sum & Difference Formulas

NAME	FORMULA	NOTES
Sine Sum	$\sin(A + B) = \sin A \cos B + \cos A \sin B$	
Sine Diff	$\sin(A - B) = \sin A \cos B - \cos A \sin B$	
Cosine Sum	$\cos(A + B) = \cos A \cos B - \sin A \sin B$	
Cosine Diff	$\cos(A - B) = \cos A \cos B + \sin A \sin B$	
Tangent Sum	$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$	

Co-Function Identities

NAME	FORMULA	NOTES
Sine-Cosine	$\sin\left(\frac{\pi}{2} - \theta\right) = \cos(\theta)$	
Cosine-Sine	$\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$	
Tangent-Cotangent	$\tan\left(\frac{\pi}{2} - \theta\right) = \cot(\theta)$	

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